BRAC UNIVERSITY

Principles of Physics-II (PHY-112)

Department of Mathematics and Natural Sciences

Assignment: 02 — **Section**: 02

Dispatch Date: April 13, 2025

Submission Deadline: April 19, 2025

Duration: 6 Days | **Spring 2025 (10F-29C)** | Marks: 30

Assignment Instructions

- Write your assignment on A4-sized paper. Use SI units. Properly staple and organize your pages.
- Attempt the questions set for your designated section only. Follow the deadline as written above.
- Ensure your solutions are clear, concise, and well-organized. Add diagrams where needed. Expand your answers when necessary. Do not spit out equations with no context. Steps hold marks.
- **Plagiarism, cheating, copying someone else's script, or the blatant use of AI tools** will result in a 50% decrement or a zero score in extreme cases.
- Late submissions will be penalized at a rate of **10% per day** for two days. No submissions will be accepted two days after the deadline. Especially after the solution is posted. Online submission is not an option.

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1. A Geiger tube comprises two very long, coaxial cylindrical parts. A thin wire is in the center, and a hollow cylinder is around the other. The central wire is positively charged, and the outer cylinder is negatively charged. The potential difference between the wire and the cylinder is 2 kV.
   (a) What is the direction of the electric field inside the tube? Which element is at a higher electric potential? What is (are) the shape(s) of the equipotential surfaces inside the tube? Draw a diagram that shows the cross-section of the tube from the front. **(4)**
   (b) Consider two equipotential surfaces described in (a). Suppose they differ in electric potential by 15 V. Do two such equipotential surfaces near the central wire have the same spacing as they would near the outer cylinder? If not, where in the tube are the equipotential surfaces that are more widely spaced? Explain your answer with the same diagram you drew in (a). **(5)**

2. Consider two parallel uniformly charged plates that are equal but oppositely charged. (i) What is (are) the shape(s) of the equipotential surfaces in the region between them? Explain your answer. (ii) What is (are) the shape(s) of the equipotential surfaces in the regions on either side of them? Explain your answer with a diagram. **(4)**

3. The electric field on the y-axis due to a point charge fixed at the origin is given by $\vec{E} = -\frac{a}{y^3}\hat{j}$ where $a = 10\,\text{kV m}$ and $y \neq 0$. (i) Find the magnitude and sign of the point charge. (ii) Find the potential difference between the points on the y-axis at $y = 1.0\,\text{m}$ and $y = 2.0\,\text{m}$. Which of these points has a higher potential? Drawing a diagram is mandatory. **(6)**

4. One non-conducting rubber ball and one conducting metal ball, each of radius 2.5 cm, are placed 1.5 m apart from center to center. Ball 1 has a charge of $+2.0 \times 10^{-8}\,\text{C}$, and ball 2 has a charge of $-1.0 \times 10^{-8}\,\text{C}$. The balls are far enough apart that they don't affect each other's charge distribution. Take the electric potential to be zero at infinity. (i) What is the electric potential at the point exactly halfway between the centers of the two balls? (ii) What is the net potential 0.5 cm inside the surface of (ii) ball 1 and (iii) ball 2, respectively, one at a time? **(6)**

5. While building an electric scooter, you need to temporarily splice, end to end, a 1.2 m iron wire with an 8.0 m nickel wire that has the same diameter. The maximum current in the wires is 1.5 A. (i) Find the (i) resistances and (ii) potential drop across each wire of this system when the current is 1.5 A. (iii) Find the electric field in each wire. Where does this electric field come from? Is it uniform or non-uniform? **(5)**

Important Constants and Order of Magnitude Prefixes	
Constants	**Prefixes**
Vacuum Permittivity, ϵ_0 8.854×10^{-12} F m^{-1}	10^{-15} = f (femto) 10^{-12} = p (pico)
Vacuum Permeability, μ_0 $4\pi \times 10^{-7}$ T m A^{-1}	10^{-9} = n (nano) 10^{-6} = µ (micro)
Speed of Light, c 3×10^{8} m s^{-1}	10^{-3} = m (milli) 10^{-2} = m (centi)
Electron Charge, e 1.602×10^{-19} C	10^{2} = h (hecto) 10^{3} = k (kilo)
Proton mass, m_p 1.673×10^{-27} kg	10^{6} = M (mega) 10^{9} = G (giga)
Electron mass, m_e 9.109×10^{-31} kg	10^{12} = T (tera) 10^{15} = P (peta)

May the Force be with you!